

PROJECT N°4

Secure Internal Network Administration

ALMASSAR Enterprise Infrastructure

Company:ALMASSAR

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1. Introduction and Objectives

The objective of this project is to design, deploy, and administer a secure internal network infrastructure for the ALMASSAR company. The company is divided into three distinct operational departments: Purchases, Clients, and Human Resources.

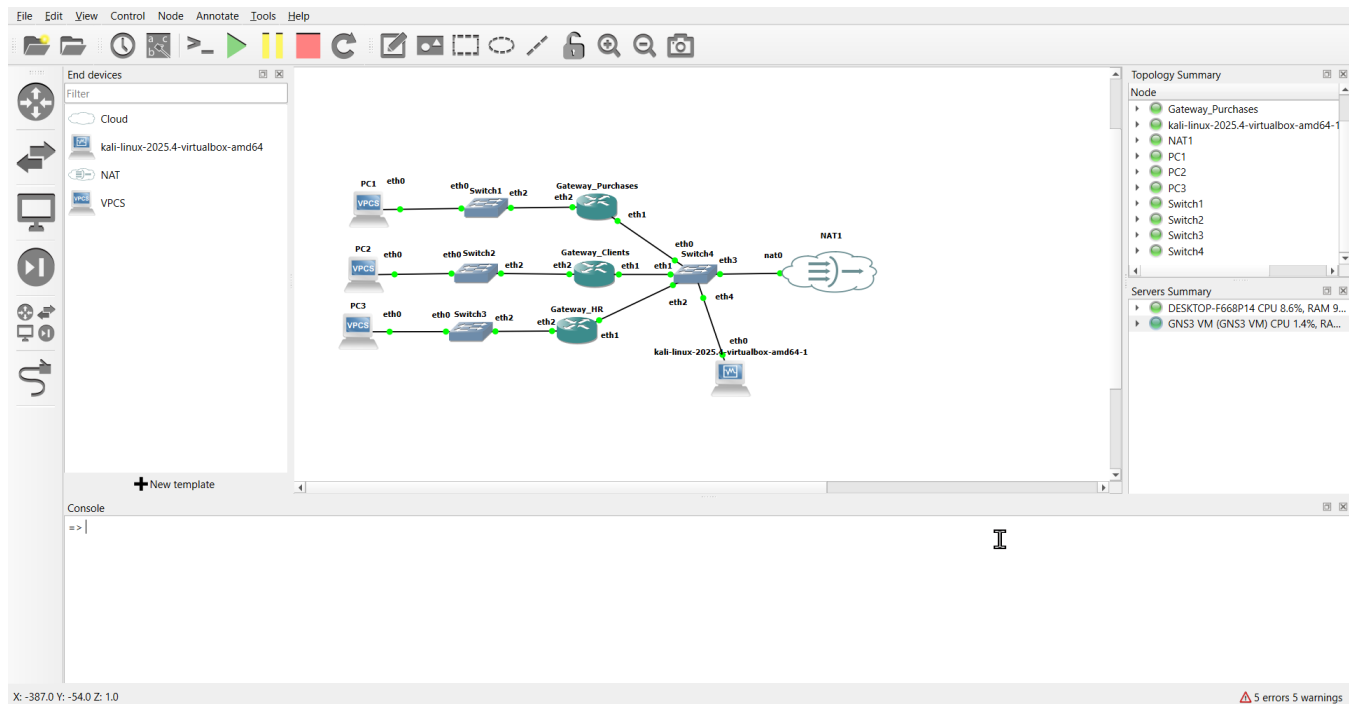
To ensure both high performance and robust security, the network architecture isolates these departments into separate logical segments while allowing controlled inter-departmental communication via a core backbone. Furthermore, the infrastructure provides essential enterprise services, including DHCP, DNS, NAT, HTTPS, and FTPS, ensuring secure internal and external data flow.

2. Network Topology and Addressing

The network was deployed using the GNS3 simulation platform. A central backbone network connects distinct departmental gateways, allowing routed communication while enforcing security boundaries.

The IP addressing scheme is organized as follows:

| Department                       | IP Subnet       |
|----------------------------------|-----------------|
| Purchases Department             | 172.17.1.0/24   |
| Clients Department               | 172.17.4.0/24   |
| Human Resources Department       | 172.17.6.0/24   |
| Enterprise Services (Kali Linux) | 192.168.42.0/24 |



### 3. Enterprise Services Implementation

#### 3.1 Dynamic Host Configuration Protocol (DHCP)

To support automated IP address assignment for end-user devices, a centralized DHCP server was configured on the Kali Linux machine. The MikroTik gateways were configured as DHCP relays to forward broadcast requests from the distinct departmental subnets to the central server.

```

Gateway_P Gateway_C Gateway_H PC1 PC2 PC3 PC1
PC1> ip dhcp
DORA IP 172.17.1.50/24 GW 172.17.1.1

```

#### 3.2 Domain Name System (DNS)

A local name resolution service was implemented using BIND9 to allow users to access internal resources using human-readable domain names rather than IP addresses. The enterprise domain almassar.uir.ma was created, with forward lookup zones resolving requests to the central server at 192.168.42.48.

```

PC1> ping www.almassar.uir.ma
www.almassar.uir.ma resolved to 192.168.42.48

84 bytes from 192.168.42.48 icmp_seq=1 ttl=63 time=1.313 ms
84 bytes from 192.168.42.48 icmp_seq=2 ttl=63 time=1.708 ms
84 bytes from 192.168.42.48 icmp_seq=3 ttl=63 time=1.596 ms
^C

```

#### 3.3 Secure Web Server (HTTPS)

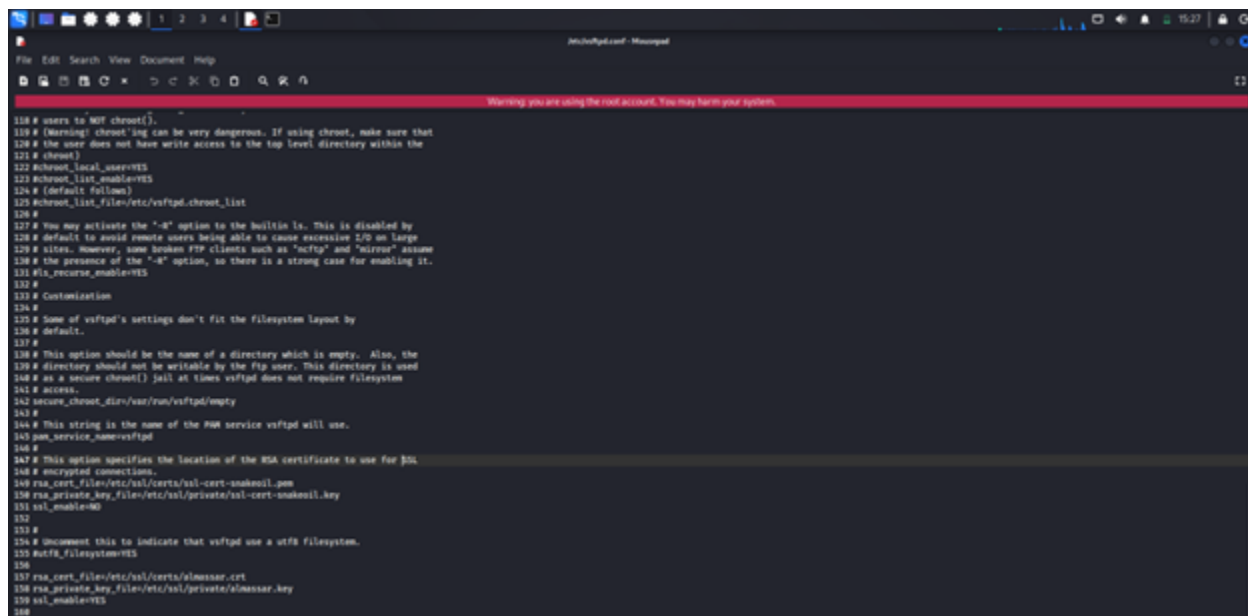
To host ALMASSAR's internal web applications securely, an Apache2 web server was deployed. To meet cybersecurity requirements, port 80 (HTTP) was secured by generating a custom self-signed X.509 RSA certificate specifically for the ALMASSAR organization, ensuring that all traffic is encrypted via TLS.

```
(kali@kali)~$ echo | openssl s_client -connect localhost:443 -brief
Connecting to ::1
Can't use SSL_get_servername
depth=0 C=mc, ST=rabat, L=sidi bouknadel, O=almassar, OU=www.almassar.uir.ma, CN=www.almassar.uir.ma, emailAddress=suiveursoul@gmail.com
verify error:num=10:self-signed certificate
CONNECTION ESTABLISHED
Protocol version: TLSv1.3
Cipher suite: TLS_AES_256_GCM_SHA384
Peer certificate: C=mc, ST=rabat, L=sidi bouknadel, O=almassar, OU=www.almassar.uir.ma, CN=www.almassar.uir.ma, emailAddress=suiveursoul@gmail.com
Hash used: SHA256
Signature type: rsa_pss_rsae_sha256
Verification error: self-signed certificate
Negotiated TLS1.3 group: X25519MLKEM768
DONE
```

### 3.4 Secure File Transfer (FTPS)

To facilitate the secure exchange of documents across the enterprise, the vsftpd service was installed. The configuration was hardened by disabling anonymous access and enforcing SSL/TLS encryption using the previously generated ALMASSAR certificate, thereby preventing packet sniffing of credentials and data.

```
(kali@kali)~$ sudo mousepad /etc/vsftpd.conf
```



```
Warning: you are using the root account. You may harm your system.

118 # users to NOD chroot().
119 # (Warning: chroot'ing can be very dangerous. If using chroot, make sure that
120 # the user does not have write access to the top level directory within the
121 # chroot)
122 #chroot_local_users=YES
123 #chroot_list_enable=YES
124 # (default follows)
125 #chroot_list_file=/etc/vsftpd.chroot_list
126 #
127 # You may activate the "-d" option to the builtin ls. This is disabled by
128 # default to avoid remote users being able to cause excessive I/O on large
129 # files. However, some broken FTP clients such as "ncftp" and "Mirrorm" assume
130 # the presence of the "-d" option, so there is a strong case for enabling it.
131 #ls_recurs_enable=YES
132 #
133 # Customization
134 #
135 # Some of vsftpd's settings don't fit the filesystem layout by
136 # default.
137 #
138 # This option should be the name of a directory which is empty. Also, the
139 # directory should not be writable by the ftp user. This directory is used
140 # as a secure chroot() jail at times vsftpd does not require filesystem
141 # access.
142 #secure_chroot_dir=/var/run/vsftpd/empty
143 #
144 # This string is the name of the PAM service vsftpd will use.
145 #pam_service_name=vsftpd
146 #
147 # This option specifies the location of the RSA certificate to use for SSL
148 # encrypted connections.
149 #rsa_cert_file=/etc/ssl/certs/ssl-cert-snakeoil.pem
150 #rsa_private_key_file=/etc/ssl/private/ssl-cert-snakeoil.key
151 #ssl_enable=NO
152 #
153 #
154 # Uncomment this to indicate that vsftpd use a utf8 filesystem.
155 #utf8_filesystem=YES
156 #
157 #rsa_cert_file=/etc/ssl/certs/almassar.crt
158 #rsa_private_key_file=/etc/ssl/private/almassar.key
159 #ssl_enable=YES
160 #
```

```
157 rsa_cert_file=/etc/ssl/certs/almassar.crt
158 rsa_private_key_file=/etc/ssl/private/almassar.key
159 ssl_enable=YES
```

## 4. Network Security and Routing

### 4.1 Departmental Isolation (Firewall Filtering)

A core requirement of the ALMASSAR network is that communication between departments must be strictly controlled. To implement a Zero Trust model, firewall rules were applied to the MikroTik forward chains.

### Example Rule

A rule was established to explicitly drop all ICMP traffic originating from the Purchases department (172.17.1.0/24) destined for the Human Resources department (172.17.6.0/24).

```
PC1> ping 172.17.6.50
172.17.6.50 icmp_seq=1 timeout
172.17.6.50 icmp_seq=2 timeout
172.17.6.50 icmp_seq=3 timeout
^C
PC1> ping www.almassar.uir.ma
www.almassar.uir.ma resolved to 192.168.42.48
84 bytes from 192.168.42.48 icmp_seq=1 ttl=63 time=1.188 ms
84 bytes from 192.168.42.48 icmp_seq=2 ttl=63 time=1.545 ms
84 bytes from 192.168.42.48 icmp_seq=3 ttl=63 time=1.459 ms
^C
```

## 4.2 Network Address Translation (NAT) & Internet Access

To provide the internal users with controlled access to the external Internet, an IP Masquerade rule was configured on the edge router. This allows the private departmental IP addresses to be translated into a publicly routable IP address.

```
PC1> ping 8.8.8.8
84 bytes from 8.8.8.8 icmp_seq=1 ttl=253 time=23.977 ms
84 bytes from 8.8.8.8 icmp_seq=2 ttl=253 time=22.329 ms
84 bytes from 8.8.8.8 icmp_seq=3 ttl=253 time=24.666 ms
84 bytes from 8.8.8.8 icmp_seq=4 ttl=253 time=35.769 ms
^C
PC2> ping 8.8.8.8
84 bytes from 8.8.8.8 icmp_seq=1 ttl=253 time=22.092 ms
84 bytes from 8.8.8.8 icmp_seq=2 ttl=253 time=22.327 ms
84 bytes from 8.8.8.8 icmp_seq=3 ttl=253 time=23.032 ms
^C
```

## 5. Conclusion

The ALMASSAR enterprise network infrastructure has been successfully modeled, deployed, and secured according to the organizational requirements. The logical separation of departments via routed backbone architecture significantly reduces broadcast domains and improves overall performance.

Furthermore, the successful integration of hardened enterprise services—including centralized DHCP, DNS, HTTPS, and FTPS—alongside strict firewall and NAT policies, ensures a resilient, scalable, and highly secure environment for daily corporate operations.

All project objectives have been met. The infrastructure demonstrates a production-grade approach to enterprise network security with departmental isolation, authenticated services, and encrypted communications.